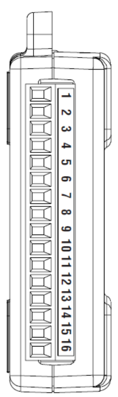
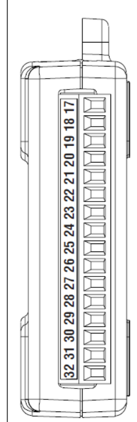
All channels are accessible via removable screw terminals, divided between analog and digital terminals, which allow a quick and flexible connection to signal cables (from 16 to 28 AWG).

The device supports:

- Analog acquisition modes:
  - Single-ended (RSE): all channels measure with respect to GND.
  - Differential (DIFF): measurements on channel pairs, with greater noise immunity.
- External digital trigger (via PFI 0) for measurement synchronization with external events.
- Event counter for counting logic transitions (up to 5 MHz).
- Terminal Assignments

Module	Terminal	Signal, Single-Ended Mode	Signal, Differential Mode
	1	GND	GND
	2	AI 0	AI 0+
	3	AI 4	AI 0–
	4	GND	GND
	5	AI 1	AI 1+
	6	AI 5	AI 1–
	7	GND	GND
	8	AI 2	AI 2+
	9	AI 6	AI 2–
	10	GND	GND
	11	AI 3	AI 3+
	12	AI 7	AI 3–
	13	GND	GND
	14	AO 0	AO 0
	15	AO 1	AO 1
	16	GND	GND

(a) Analog Terminal Assignments

Module	Terminal	Signal
	17	P0.0
	18	P0.1
	19	P0.2
	20	P0.3
	21	P0.4
	22	P0.5
	23	P0 6
	24	P0.7
	25	P1.0
	26	P1.1
	27	P1.2
	28	P1.3
	29	PFI 0
	30	+2.5 V
	31	+5 V
	32	GND

(b) Digital Terminal Assignments

Figure 5

Created in LABVIEW 2021

“Programmer’s Guide”

Course: Measurements for Automation and Industrial Production

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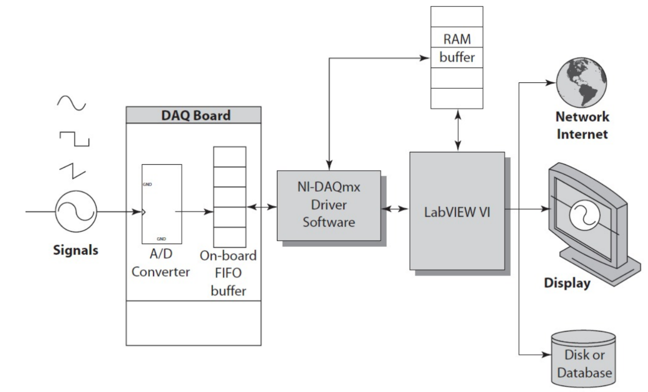


Figure 4

The capabilities of the NI USB-6008 include:

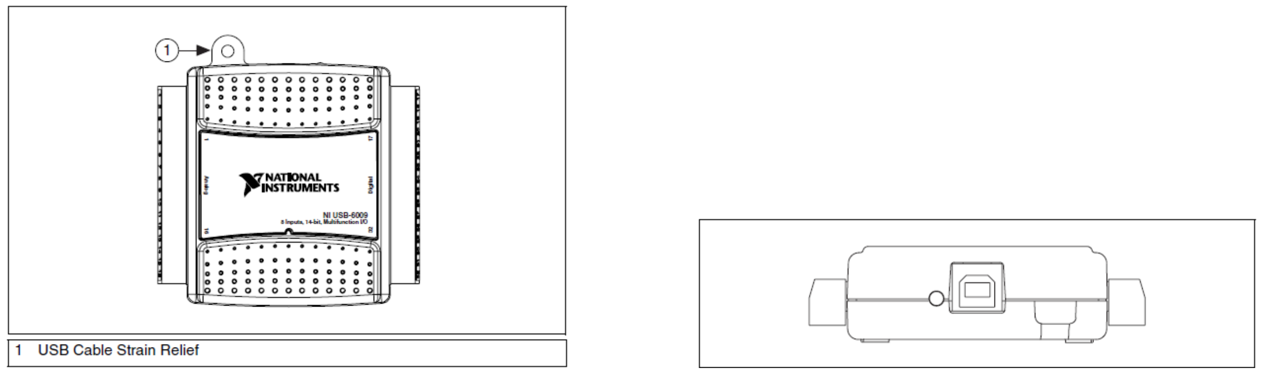
- **Analog Input:** Acquisition of analog signals from external sensors or devices.
- **Analog Output:** Generation of analog signals for driving actuators or simulating processes.
- **Digital I/O:** Reading and writing of TTL digital signals, crucial for control and logic applications.
- **Counter/Timers:** Counting of digital events using a high-resolution 32-bit counter.
- **Power Sourcing:** Provision of regulated +5V and +2.5V outputs to power external low-power devices.

Internal Architecture

The device incorporates:

- A 12-bit ADC that, through a multiplexer (MUX) and a programmable gain amplifier (PGA), allows to selection and acquisition of one of the analog channels in differential or single-ended mode;
- two 12-bit DACs to generate 0–5 V analog signals;
- an integrated USB microcontroller, which manages data transfer, timing and control logic;
- an internal FIFO for data buffering during acquisition.

NI DAQ USB-6008 data acquisition



(a) USB-6008 (b) USB-6008 back view

Figure 2

USB6008 Datasheet[1]

The NI USB-6008 [9] provides connection to eight analog input (AI) channels, two analog output (AO) channels, 12 digital input/output (DIO) channels, and a 32-bit counter with a full-speed USB interface.

The following block diagram shows key functional components of the USB-6008:

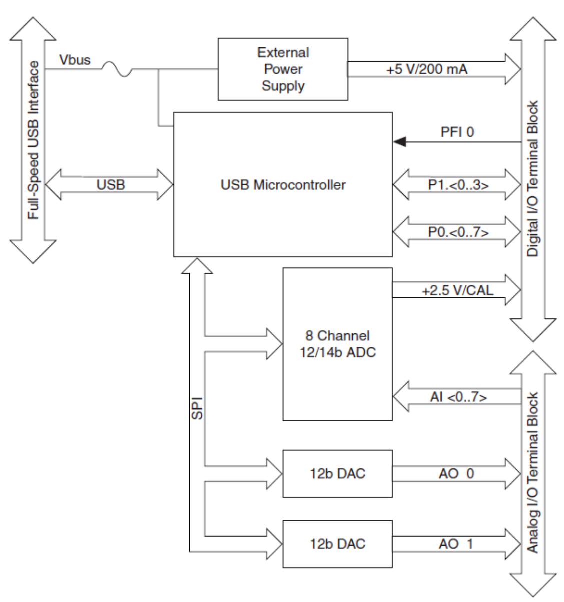


Figure 3

[1] https://www.ni.com/docs/en-US/bundle/usb-6008-specs/page/specs.html?srsltid=AfmBOoomX5uQUUn1iMTOb3gol5ElTAT813--B6ktlUhKk4vixIqbCJly

Abstract

This document provides the necessary background information on the calibration methods and the software applications used, with the ultimate goal of producing a guide for the programmers who wish to either modify or extend the capabilities of the software suite developed. This guide is an attempt at making life easier for the user by describing how each step in the calibration procedure has been implemented and how each device connection has been handled. The project's aim is to develop an automated measurement system for characterizing the Renesas FS-1012-1100-NG transducer. This thermal mass flow transducer's output is a voltage signal. This signal has to be analysed and compared with the measurement produced by a pre-calibrated transducer of similar characteristics. For this aim, the Renesas FS-2012-1100-NG was chosen. The task involves several steps, which are all dealt with using a unique software application developed within the LabVIEW development environment: the main VI. Nevertheless, this program calls several other sub-routines to leverage multithreading and parallel programming to handle multiple device connections simultaneously, and to manage each user request in response to a GUI event. Finally, the project's main software application includes a set of data analysis tools, which bring up the statistical distribution of samples in each data set acquired, reveal the presence of Hysteresis, and compute the combined standard uncertainty resulting from a set of uncertainty sources within the measuring system itself.

Chapter I

Programmable Hardware Instrumentation Used

Power Supply Agilent E36436A

The Agilent E3633A/E3634A DC Power Supply [4] was configured for Constant Voltage (CV) operation, with the output voltage fixed at 12V.

Figure 1: Power Supply E3634A

Main characteristics

- **Voltage:** ±(% of setting + offset)
  - **Low Range (0 to 8 V):** ±(0.05% + 5 mV)
  - **High Range (0 to 20 V):** ±(0.05% + 15 mV)
- **Current:** ±(% of setting + offset)
  - **Low Range (0 to 20 A):** ±(0.2% + 5 mA)
  - **High Range (0 to 10 A):** ±(0.2% + 5 mA)